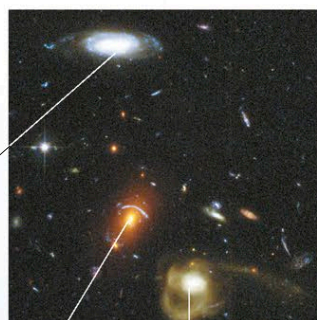


## TIME AND EXPANDING SPACE

### PEERING INTO DEEP SPACE

This Hubble “deep-field” photograph shows a jumble of galaxies viewed at different distances. Each appears as it existed billions of years ago.



diffuse, young galaxy not yet condensed into a tight spiral

young, blue galaxy 4 billion light-years away, pictured as it was 4 billion years ago

elliptical galaxy, 6 billion light-years away

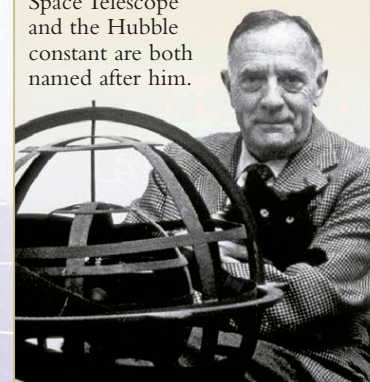
spiral galaxy, 3 billion light-years distant

The continued expansion of space, combined with the constant speed of light, turns the universe into a giant time machine. The light from a remote galaxy has taken billions of years to reach Earth, so astronomers see the galaxy as it was billions of years ago. In effect, the deeper astronomers look into space, the farther they peer into the universe’s history. In the remotest regions, they see only incompletely formed galaxies as they looked soon after the Big Bang. The most dim and distant galaxies are now receding from Earth at a rate faster than the speed of light. We can still see light that was emitted from them in the distant past, but eventually they will fade from view.

At greater distances yet, beyond the observable universe (see p.23), there may exist other objects that have moved away so fast that light from them has never reached Earth.

### EDWIN HUBBLE

American astronomer Edwin Hubble (1889–1953) is famous for being the first to prove that the universe is expanding, though several other astronomers—such as the Belgian Georges Lemaître—made important contributions to this discovery. Hubble is also noted for his earlier proof that galaxies are external to the Milky Way and for his system of galaxy classification. The Hubble Space Telescope and the Hubble constant are both named after him.



## LOOKBACK DISTANCE

The expansion of space complicates the expression of distances to very remote objects, particularly those that we now observe as they existed more than 5 billion years ago. When astronomers describe the distance to such faraway objects, by convention they use the “lookback” or “light-travel-time” distance. This is the distance that light from the object has traveled through space to reach us today, and it tells us how long ago the light left the object. But because space has expanded in the interim, the distance of the galaxy when the light began its journey toward Earth is less than the lookback distance. Conversely, the true distance to the remote object (called the “comoving” distance) is greater than the lookback distance. These distinctions need to be remembered when, for example, a galaxy is stated as being 10 billion light-years away.

### DIVERGING WORLDS

An object described as being 11 billion light-years away (lookback distance) has a greater true distance (comoving distance) due to the effects of the universe’s expansion.

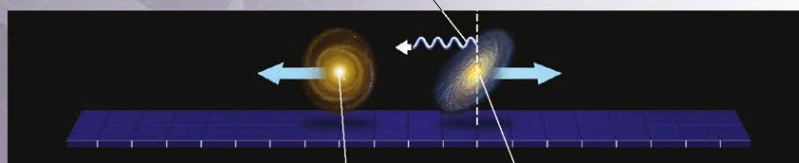
1. Eleven billion years ago, a photon of light departs distant galaxy X traveling toward the Milky Way. The two galaxies are separated by 4 billion light-years of space.

2. Six billion years later, the photon has not yet reached its destination, because space has expanded, carrying the galaxies much farther apart.

voids between galaxy clusters progressively enlarge and become almost empty of dust and gas

3. The photon reaches the Milky Way, where an observer sees X as it was 11 billion years ago, 11 billion light-years away (lookback distance). Meanwhile, X’s true (comoving) distance has increased to 18 billion light-years.

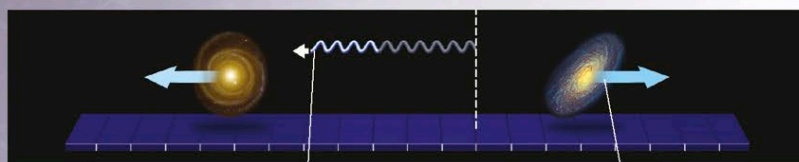
photon leaves galaxy X



11 BILLION YEARS AGO

Milky Way

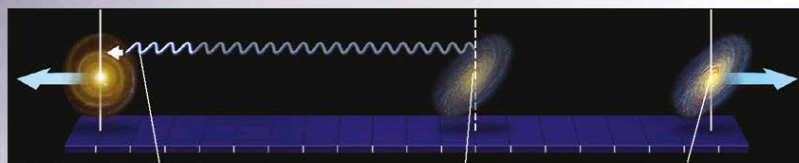
distant galaxy X receding



5 BILLION YEARS AGO

photon travels toward Milky Way

galaxy X still receding



PRESENT DAY

photon arrives

lookback distance

true, comoving distance

### ACCELERATING EXPANSION

This is a conceptual interpretation of how a region of space may have changed over a 9-billion-year period. As space has expanded, the galaxies within it have been carried apart, evolving as they go. This interpretation shows expansion speeding up—a scenario now accepted by most cosmologists.